

IMPROVED KISSING NUMBERS IN NINETEEN AND TWENTY-ONE DIMENSIONS

ABSTRACT. We prove that the kissing numbers in 19 and 21 dimensions are at least 12268 and 30761. The previous records were 11948, due to Ho, and 29768, due to Cohn and Li, so the gains are 320 and 993. Both constructions start from the odd-sign kissing configurations of Cohn and Li, delete a small block of their points, and add more new points than were deleted. In 19 dimensions we cut the 20-dimensional configuration by the hyperplane orthogonal to the indicator vector of a tetrad, project 20-dimensional sign vectors into that hyperplane, and give their tails one of two lengths. The proof is a case analysis. In 21 dimensions the new points are weighted sign vectors on six cosets of a $[21, 9, 6]$ code. They are given by an explicit table of 384 integer vectors, and the proof is an exact computation, reduced by a sign symmetry of order 8. We have also formalized both results in Lean 4 with Mathlib.

1. INTRODUCTION

The kissing problem in $\mathbb{R}^n$ asks how many unit spheres can touch a central unit sphere without overlapping [2, Ch. 1, §2.2]. In terms of the points of tangency, it asks for the largest set X of unit vectors in $\mathbb{R}^n$ with $\langle x, y \rangle \leq 1/2$ for all distinct $x, y \in X$. We write τ_n for this maximum, as in [2, Ch. 13]. It is known only for $n \in \{1, 2, 3, 4, 8, 24\}$; see [1, Table 1.1] for references.

For $17 \leq n \leq 23$ the best constructions were Leech’s cross sections of the Leech lattice minimal vectors [6] until Cohn and Li [1] improved dimensions 17 through 21. In 19, 20 and 21 dimensions their idea is short. Take the classical configuration built from a constant-weight code [7], and put an odd number of minus signs on each codeword instead of an even number. Every sign vector of a code orthogonal to the constant-weight code then has inner product at most 6 with the old points, while the kissing condition allows $\frac{1}{2}\sqrt{8n}$, which is larger for $n \geq 19$. So these sign vectors can be added, one per word of a code with minimum distance at least $n/4$. Cohn and Li obtained $\tau_{19} \geq 11692$ in this way, and Ho [5] found a larger code of 1280 words for the same purpose, reaching $\tau_{19} \geq 11948$.

Theorem 1.1. *We have $\tau_{19} \geq 12268$ and $\tau_{21} \geq 30761$.*

Table 1 compares these values with the previous records and the known upper bounds.

The mechanism is the slack between 6 and $\frac{1}{2}\sqrt{8n}$. Cohn and Li’s added points are flat: each is a sign vector scaled to squared norm 8, and none of them touches an old point. We spend this unused room by changing the shape of the added points.

TABLE 1. Kissing number bounds in dimensions 19 and 21. The previous lower bounds are due to Ho [5] in dimension 19 and to Cohn and Li [1] in dimension 21.

dimension	previous lower bound	new lower bound	upper bound
19	11948 [5]	12268	24417 [3]
21	29768 [1]	30761	53524 [3]

- In 19 dimensions (Section 3) the added points come from 20 dimensions. Projecting them into a hyperplane leaves a short three-dimensional tail, and we multiply it by 9/11 or 7/5 according to its direction. This lets two added points differ in only four head coordinates when their tails differ, which a code of minimum distance at least 5, such as Ho's, must avoid. We add 1792 points where Ho adds 1280. The long tails come too close to 192 old points, which we delete, for a net gain of $1792 - 192 - 1280 = 320$.
- In 21 dimensions (Section 4) each added point carries its own positive weight on each coordinate. That freedom allows six cosets of a $[21, 9, 6]$ code, three even and three odd, where Cohn and Li use only the four odd cosets of the same code. The weights overshoot one octad, and we delete 31 points on it, for a net gain of $3072 - 31 - 2048 = 993$.

The two proofs differ in character. The 19-dimensional proof is a case analysis that can be checked by hand, given the finite facts about a code of size 4096 stated in Lemma 3.1. The 21-dimensional proof depends on computer calculation in an essential way, unlike the proofs in [1]. Its added points are integer vectors with entries between $1.4 \cdot 10^8$ and $2.5 \cdot 10^8$ in absolute value. They were obtained by numerical optimization and rounding, not by hand. The theorem is a statement about that explicit table, and its proof is a list of about 11 million exact integer inequalities. We checked these with two independent programs, and we also formalized both results in Lean 4 with Mathlib (Section 5). The formal proof does not verify our case analysis. It verifies the conclusion, that explicit sets of 12268 and 30761 unit vectors with pairwise inner products at most $1/2$ exist.

2. THE ODD-SIGN CONFIGURATIONS

We identify a binary word with its support. For $c \in \mathbb{F}_2^n$ we write $\text{wt}(c)$ for its weight, $(-1)^c \in \{\pm 1\}^n$ for its sign vector, and $\lambda \circ q$ for the coordinatewise product of two vectors.

Let $\mathcal{O}$ be a set of words of length n with weight 8 and pairwise intersections of size at most 4. The *odd configuration* $B(\mathcal{O})$ consists of the $4\binom{n}{2}$ roots $\pm 2e_i \pm 2e_j$ and, for each $o \in \mathcal{O}$, the 128 vectors with entries ± 1 on o , zeros elsewhere and an odd number of minus signs [1, §2]. All have squared norm 8.

Lemma 2.1 (Cohn and Li). *Distinct $b, b' \in B(\mathcal{O})$ satisfy $\langle b, b' \rangle \leq 4$.*

Proof. Two distinct roots have inner product in $\{0, \pm 4, -8\}$. A root has two nonzero coordinates, each contributing at most 2 against an octad vector, so that inner product is at most 4. Two vectors on the same octad have odd sign patterns, so they differ in an even and positive number $d \geq 2$ of places and have inner product $8 - 2d \leq 4$. Two vectors on different octads share at most four coordinates. $\square$

The next lemma is the reason for odd signs. It is the weighted form of the argument in [1, §2].

Lemma 2.2. *Let $c \in \mathbb{F}_2^n$ meet every word of $\mathcal{O}$ in an even number of coordinates, let $q = (-1)^c$, and let $\lambda \in \mathbb{R}_{\geq 0}^n$. Let b be an octad vector on $o \in \mathcal{O}$. Then q and b disagree in an odd number of coordinates of o , and*

$$\langle \lambda \circ q, b \rangle \leq \sum_{j \in o} \lambda_j - 2 \min_{j \in o} \lambda_j,$$

with equality for some b on o . In particular, for $\lambda = \mathbf{1}$ we have $\langle q, b \rangle = 6$ if q and b disagree in exactly one coordinate and $\langle q, b \rangle \leq 2$ otherwise. For a root $b = \pm 2e_i \pm 2e_j$ the inner product is at most $2(\lambda_i + \lambda_j)$.

Proof. The sign vector q has an even number of minus signs on o , and b has an odd number. So they disagree in an odd number of coordinates of o , in particular in at least one, and the bound follows. Equality holds for the odd pattern that agrees with q except at one coordinate where λ is smallest. For $\lambda = \mathbf{1}$, d disagreements give $\langle q, b \rangle = 8 - 2d$ with d odd. $\square$

Throughout we test compatibility in integers. For nonzero $x, y \in \mathbb{Z}^n$ the angle between them is at least 60° if and only if

$$(1) \quad \langle x, y \rangle \leq 0 \quad \text{or} \quad 4\langle x, y \rangle^2 \leq |x|^2|y|^2.$$

We call such a pair *compatible*; the same test applies to rational vectors after scaling. If every pair from a finite set of nonzero rays is compatible, then the rays normalized to unit length form a kissing configuration, and distinct rays give distinct points because (1) excludes cosine 1. With $\lambda = \mathbf{1}$, Lemma 2.2 shows that a sign vector is compatible with $B(\mathcal{O})$ when $6 \leq \frac{1}{2}\sqrt{8n}$, that is, for $n \geq 18$, with contacts when $n = 18$.

3. DIMENSION 19

3.1. The codes. Coordinates are $1, \dots, 20$. Let $E \subseteq \mathbb{F}_2^{20}$ be the span of the twelve words

$$\begin{aligned} &\{1, 8, 9, 12, 16, 17, 18, 19\}, \{2, 10, 11, 14, 15, 17, 18, 20\}, \{3, 7, 9, 13, 15, 16, 17, 19\}, \\ &\{4, 7, 8, 10, 12, 15, 16, 19\}, \{5, 10, 12, 13, 15, 16, 17, 18\}, \{6, 7, 8, 9, 10, 13, 16, 18\}, \\ &\{1, 4, 7, 9\}, \{1, 5, 6, 18, 20\}, \{1, 3, 12, 15, 20\}, \{1, 10, 13, 19, 20\}, \\ &\{1, 3, 5, 6, 7, 13, 14, 15, 18, 20\}, \{2, 4, 6, 7, 8, 13, 14, 16, 17, 18\}. \end{aligned}$$

On the first 19 coordinates these are Ho's generators $m_1, \dots, m_6, s_1, \dots, s_4, r_1, r_2$ [5, §2]. Our coordinate 20 is Ho's coordinate 24, the last column of his extension matrix P , so we have added it to the words where that column has a 1. Ho shows that the full extension to length 24 is the extended binary Golay code [5, Prop. 2]. So E is the Golay code punctured at Ho's coordinates 20 to 23, and its dual $E^\perp$ is the Golay code shortened at the same four coordinates [8]. The 130 words of weight 8 in $E^\perp$ are the *octads*. They form the shortened Steiner system $S(5, 8, 24)$ with $n = 20$ of [1, Table 2.1], so $B(\text{octads})$ is Cohn and Li's 17400-point configuration C_{20} in $\mathbb{R}^{20}$.

The five octads of the Golay code that contain four given coordinates partition the other twenty coordinates into five tetrads, and these six tetrads form a *sextet* [2, Ch. 10]. Puncturing at the four given coordinates turns those five octads into five words of weight 4 in E .

The following facts are finite computations on the 4096 words of E .

Lemma 3.1. (1) *The weight enumerator of E is*

$$\begin{aligned} &1 + 5z^4 + 64z^5 + 240z^6 + 320z^7 + 250z^8 + 640z^9 + 1056z^{10} \\ &+ 640z^{11} + 250z^{12} + 320z^{13} + 240z^{14} + 64z^{15} + 5z^{16} + z^{20}. \end{aligned}$$

The five words of weight 4 are pairwise disjoint and cover all 20 coordinates.

- (2) *Let $T = \{1, 4, 7, 9\}$, one of the five, and let H be the other 16 coordinates, the head. For $e \in E$ write e_T and e_H for its parts on T and H . The nonzero words with $\text{wt}(e) \leq 7$ have $(\text{wt}(e_T), \text{wt}(e_H))$ equal to $(4, 0)$ for 1 word, $(0, 4)$ for 4, $(1, 4)$ for 64, $(0, 6)$ for 96, $(2, 4)$ for 144, $(1, 6)$ for 256 and $(3, 4)$ for 64.*
- (3) *Two distinct octads meet in at most 4 coordinates. The octads meet T in 0, 2 or 4 coordinates, for 30, 96 and 4 octads respectively.*

The intersections in (3) are even because $T \in E$ and the octads lie in $E^\perp$. By (2), two distinct words $c, c' \in E$ with $d_T = \text{wt}(c_T + c'_T)$ and $d_H = \text{wt}(c_H + c'_H)$ satisfy the following. If $d_T = 0$, then $d_H = 4$ (and $c + c'$ is a tetrad inside H) or $d_H \geq 6$. If $1 \leq d_T \leq 3$, then $d_H \geq 4$. If $d_T = 4$, then $d_H = 0$ (and $c' = c + \mathbf{1}_T$), $d_H = 4$ or $d_H \geq 6$.

3.2. The configuration. Let

$$R = \begin{pmatrix} 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix},$$

acting on the coordinates $(1, 4, 7, 9)$ of T . The rows of $R/2$ are an orthonormal basis of $\mathbf{1}_T^\perp \cap \mathbb{R}^T$, so $(R/2)^\top(R/2) = I - J/4$. The map $\varphi(v) = (v_H, Rv_T/2) \in \mathbb{R}^{16} \times \mathbb{R}^3$ therefore satisfies

$$(2) \quad \langle \varphi(u), \varphi(v) \rangle = \langle u, v \rangle - \frac{1}{4} \left(\sum_{j \in T} u_j \right) \left(\sum_{j \in T} v_j \right),$$

and it is an isometry from the hyperplane $\mathbf{1}_T^\perp$ onto $\mathbb{R}^{19}$. The configuration in $\mathbb{R}^{19}$ has two parts.

Base. There are 10668 points of C_{20} orthogonal to $\mathbf{1}_T$, as many as in Leech's 19-dimensional configuration. We delete from them the 192 points that lie on the four octads containing T ; these are the points with two plus and two minus signs on T . The remaining 10476 points b enter as $\varphi(b)$. They are the 480 roots inside H , the 12 roots $\pm 2(e_i - e_j)$ with $i, j \in T$, the 3840 odd vectors on the 30 octads disjoint from T , and the 6144 odd vectors on the 96 octads meeting T twice whose two T -entries have opposite signs.

Additions. Let $c \in E$ with $\text{wt}(c_T) \in \{1, 2, 3\}$ and $q = (-1)^c$. The projected tail $t_0 = Rq_T/2$ satisfies $|t_0|^2 = 4 - \frac{1}{4}(\sum_T q_j)^2$ by (2). When $\text{wt}(c_T)$ is odd, $t_0 \in \{\pm 1\}^3$ and $|t_0|^2 = 3$; we call c *diagonal*. When $\text{wt}(c_T) = 2$, $t_0 = \pm 2e_k$ and $|t_0|^2 = 4$; we call c an *axis* word. The added point is

$$a(c) = (q_H, \alpha t_0), \quad \alpha = \begin{cases} 9/11, & c \text{ diagonal,} \\ 7/5, & c \text{ axis,} \end{cases}$$

with $|a(c)|^2 = 2179/121$ or $596/25$. Multiplying by 22 or 10 makes it an integer vector, and the base points $\varphi(b)$ become integer vectors after multiplying by 2.

Which words. Let W be the span of the five tetrads, a code of size 32. The 3584 words with $\text{wt}(c_T) \in \{1, 2, 3\}$ fall into 112 cosets of W . Each coset $c + W$ is a 5-cube: its vertices are the 32 words, and each of the five directions adds one tetrad. We keep one of the two bipartition classes of each cube, 16 words. Lemma 3.2 below holds for either class, so any choice gives a valid configuration of the same size. For definiteness, and to match `data/D19_12268_rays.txt`, we keep the words c for which $c + c_{\min}$ is a sum of an even number of tetrads, where $c_{\min}$ is the smallest word of the coset read as the integer $\sum_{j \in c} 2^{j-1}$. This gives 1024 diagonal and 768 axis words, 1792 in all.

The total is $10668 - 192 + 1792 = 12268$. Table 2 lists the blocks.

3.3. Proof. Some restriction inside each cube is needed, because two words of one cube that differ by a tetrad inside H have the same T -pattern and $d_H = 4$. For two diagonal words this gives inner product $8 + 3(9/11)^2 \approx 10.01$ against the bound $\frac{1}{2}|a|^2 \approx 9.00$.

Lemma 3.2. *Two distinct kept words c, c' satisfy $c_H \neq c'_H$, and $d_H \geq 6$ if $c_T = c'_T$.*

Proof. If $c_H = c'_H$, then $c + c'$ is a word supported in T , so it is $\mathbf{1}_T$ by Lemma 3.1(2). Then c and c' lie in the same coset of W and differ by one tetrad, so they lie in opposite classes of the

TABLE 2. The blocks of the 19-dimensional configuration.

block	size	squared norm
roots in H	480	8
roots in T	12	8
octad vectors, octad disjoint from T	3840	8
octad vectors, octad meeting T twice	6144	8
diagonal additions	1024	2179/121
axis additions	768	596/25
total	12268	

TABLE 3. Largest inner product between a base point and an addition in dimension 19.

base points	count	reason	diagonal	axis
roots in H	480	$b_T = 0$	4	4
roots in T	12	$\alpha \langle q_T, b_T \rangle \leq 4\alpha$	36/11	28/5
octads disjoint from T	3840	$b_T = 0$, Lemma 2.2	6	6
octads meeting T twice	6144	three cases above	6	34/5
bound $\sqrt{2 a ^2}$			6.0014	6.9051

5-cube and are not both kept. If $c_T = c'_T$ and $d_H = 4$, then $c + c'$ is a tetrad in H , and the same argument applies. $\square$

We now check every pair, using the unnormalized vectors.

Base with base. By (2) the map φ preserves inner products on $\mathbf{1}_T^\perp$, so Lemma 2.1 gives $\langle \varphi(b), \varphi(b') \rangle \leq 4 = \frac{1}{2} \cdot 8$.

Base with additions. Since $\sum_T b_j = 0$, equation (2) gives

$$(3) \quad \langle a(c), \varphi(b) \rangle = \langle q_H, b_H \rangle + \alpha \langle q_T, b_T \rangle = \langle q, b \rangle + (\alpha - 1) \langle q_T, b_T \rangle.$$

The word c lies in E , so it meets every octad evenly, and Lemma 2.2 applies: for an octad vector b , $\langle q, b \rangle = 6$ if q and b disagree in exactly one place of the octad and $\langle q, b \rangle \leq 2$ otherwise. The bound to meet is $\frac{1}{2}|a|\sqrt{8} = \sqrt{2|a|^2}$, which is $\sqrt{4358/121} \approx 6.0014$ for diagonal and $\sqrt{1192/25} \approx 6.9051$ for axis additions.

For the 6144 points on octads meeting T twice, $b_T = \pm(e_i - e_j)$ with i, j the two coordinates of the octad in T , so $\langle q_T, b_T \rangle \in \{-2, 0, 2\}$. If it is 0, the inner product is $\langle q, b \rangle \leq 6$. If it is 2, the inner product is at most $6 + 2(\alpha - 1)$, which is $62/11$ or $34/5$. If it is -2 , then q and b disagree at both coordinates of the octad in T . The number of disagreements is odd, so it is at least 3, $\langle q, b \rangle \leq 2$, and the inner product is at most $2 - 2(\alpha - 1)$. Table 3 collects the maxima.

All maxima are below the bounds, since $36 \cdot 121 = 4356 < 4358$ and $(34/5)^2 = 1156/25 < 1192/25$.

Additions with additions. For kept words $c \neq c'$,

$$\langle a(c), a(c') \rangle = 16 - 2d_H + \alpha \alpha' \langle t_0, t'_0 \rangle, \quad \langle t_0, t'_0 \rangle = \langle q_T, q'_T \rangle - \frac{1}{4} \left(\sum_T q_j \right) \left(\sum_T q'_j \right).$$

Here $d_H \geq 4$ in every case, and $d_H \geq 6$ when $c_T = c'_T$, by Lemma 3.1(2) and Lemma 3.2. The tail product depends only on the two patterns on T :

TABLE 4. Largest inner products between two additions in dimension 19, for the cases with nonnegative tail product. The bound is $\frac{1}{2}|a||a'|$.

types	case	$\langle a, a' \rangle$	bound	cosine
diagonal, diagonal	$d_T = 0, d_H = 6$	727/121	2179/242	0.3337
	$d_T = 2, \langle t_0, t'_0 \rangle = 1, d_H = 4$	1049/121	2179/242	0.4814
axis, axis	$d_T = 0, d_H = 6$	296/25	298/25	0.4966
	$d_T = 2, d_H = 4$	8	298/25	0.3356
diagonal, axis	$d_T = 1, d_H = 4$	566/55	$\sqrt{324671/3025}$	0.4967

- two diagonal words have $d_T \in \{0, 2, 4\}$ and $\langle t_0, t'_0 \rangle = 3, \pm 1, -3$ respectively;
- two axis words have $d_T \in \{0, 2, 4\}$ and $\langle t_0, t'_0 \rangle = 4, 0, -4$ respectively;
- a diagonal and an axis word have $d_T \in \{1, 3\}$ and $\langle t_0, t'_0 \rangle = 2$ or -2 respectively.

The cases with $\langle t_0, t'_0 \rangle < 0$ give at most 8, below every bound. Table 4 lists the others at the smallest allowed d_H .

Every entry is strictly below its bound. For the diagonal–axis row we compare $(566/55)^2 = 320356/3025$ with $324671/3025$. So all pairs are compatible. The roots inside H and inside T already span $\mathbb{R}^{19}$. This proves Theorem 1.1 for $n = 19$.

3.4. Remarks. The tail lengths have little room. A diagonal addition against a full octad vector needs $36 \leq 2(16 + 3\alpha^2)$, so $\alpha \geq \sqrt{2/3} \approx 0.8165$ for diagonal words. Two axis additions with the same pattern at $d_H = 6$ need $4 + 4\alpha^2 \leq \frac{1}{2}(16 + 4\alpha^2)$, so $\alpha \leq \sqrt{2} \approx 1.4142$ for axis words. The diagonal–axis pairs at $d_T = 1, d_H = 4$ couple the two factors. Numerically, the conditions behind Tables 3 and 4 allow the diagonal factor in about $[0.8165, 0.876]$ and the axis factor in about $[1.305, 1.4142]$, in a thin region where a larger axis factor permits a larger diagonal factor. We chose $9/11$ and $7/5$ as simple fractions inside it. With plain projection, both factors equal to 1, the diagonal–axis pairs have inner product 10 against $\frac{1}{2}\sqrt{19 \cdot 20} \approx 9.75$, so plain projection fails.

Axis factors above 1 are what the 192 deleted points cannot tolerate. A deleted point has $b_T \in \{\pm 1\}^4$ with two signs of each kind. An axis word with $q_T = b_T$ whose sign vector disagrees with b in exactly one place of the octad gives inner product $6 + 4 \cdot \frac{2}{5} = 38/5 > 6.9051$. A direct check shows that each deleted point conflicts in this way with some kept axis addition.

We have not determined the largest configuration of this shape. One computed fact bears on it. In a computer screen of all 2^{16} sign vectors on H over this 10476-point base, only the 1792 heads we use admitted a compatible tail.

The additions touch nothing. By direct computation, all 2386584 contacts (pairs with cosine exactly $1/2$) are between base points. The closest pairs involving an addition have cosine $\sqrt{1089/4358} \approx 0.49989$, a diagonal addition against a full octad vector.

4. DIMENSION 21

4.1. The codes. Coordinates are $1, \dots, 24$, and the construction lives on the first 21. Let G be the extended binary Golay code: the cyclic code of length 23 generated by $g(t) = 1 + t + t^5 + t^6 + t^7 + t^9 + t^{11}$, with the coefficient of t^k at coordinate $k + 1$, extended by an overall parity bit at coordinate 24 [2, Ch. 3, §2.8]. Let $\mathcal{O}$ be the 210 octads of G that vanish on coordinates 22, 23 and 24. Then $B = B(\mathcal{O})$ is Cohn and Li’s 27720-point configuration in $\mathbb{R}^{21}$.

Let D be G punctured to the first 21 coordinates, a $[21, 12, 5]$ code. Because G is self-dual and the octads in $\mathcal{O}$ vanish on the punctured coordinates, every word of D meets every octad of $\mathcal{O}$ evenly. We call the 21 words of weight 5 in D *lines*. Two sign vectors of words of D at distance d have cosine $1 - 2d/21$, which exceeds $1/2$ only for $d = 5$, where it is $11/21$. Cohn and Li use the 2048 odd-weight words of D . The sum of two odd words is even, so no two of them are at distance 5.

Let $C \subseteq D$ be the span of the nine words (bit j of the binary expansion is coordinate $j + 1$)

1559917, 15163, 774651, 1540761, 485602, 408336, 678478, 1748752, 1231514.

It is a $[21, 9, 6]$ code with only even weights, and D/C has eight cosets. Four are even, with representatives 0, 1393, 2139, 3370, and contain no lines. Four are odd, with representatives 696, 1993, 2787, 3986, and contain 6, 3, 6 and 6 lines. These are finite computations on the 4096 words of D .

The following lemma is not used in the proof. It explains why the new points must be weighted.

Lemma 4.1. *Let c, c' be distinct words of D . If they lie in the same coset of C , or in two cosets of the same parity, then $\text{wt}(c + c') \geq 6$. If they lie in cosets of opposite parity, then $c + c'$ ranges over a whole odd coset as c and c' range over their cosets, so $\text{wt}(c + c') = 5$ occurs.*

Proof. In the first case $c + c'$ lies in C or in an even coset, and the even words of D have weight at least 6. In the second case $c + c'$ lies in an odd coset, and every odd coset contains a line. $\square$

So flat sign vectors from an even coset and an odd coset conflict at cosine $11/21$.

4.2. Weighted sign vectors. Here is the simplest way around the conflict. Let ℓ be a line, and put $\lambda_j = \sqrt{2/5}$ for $j \in \ell$ and $\lambda_j = 1$ otherwise, so $|\lambda|^2 = 18$. For $c \in D$ the point $\lambda \circ (-1)^c$ has, by Lemma 2.2, inner product at most $\sum_{j \in o} \lambda_j - 2 \min_{j \in o} \lambda_j$ with the octad vectors on o . An octad meets ℓ in 0, 2 or 4 coordinates, because $\ell \in D$, and the bound is then $8 - 2 = 6$, $6 + 2\sqrt{2/5} - 2\sqrt{2/5} = 6$, or $4 + 2\sqrt{2/5} < 6$. This is at most $\frac{1}{2}\sqrt{8}|\lambda| = 6$. So among weight vectors equal to 1 off a line and constant on it, $\sqrt{2/5}$ is the smallest weight on the line that the base allows. A small weight on ℓ weakens the conflicts at distance 5 that run through ℓ .

In general, let $c \in D$, $q = (-1)^c$ and $\lambda \in \mathbb{R}_{>0}^{21}$. Lemma 2.2 shows that $\lambda \circ q$ is compatible with every point of B exactly when

$$(4) \quad \sum_{j \in o} \lambda_j - 2 \min_{j \in o} \lambda_j \leq \sqrt{2}|\lambda| \quad \text{for all } o \in \mathcal{O}, \quad \lambda_i + \lambda_j \leq \frac{1}{\sqrt{2}}|\lambda| \quad \text{for all } i \neq j.$$

The hypothesis $c \in D$ is what makes the first condition exact. Two points $\lambda \circ q$ and $\mu \circ q'$ have inner product $\sum_j \lambda_j \mu_j (-1)^{(c+c')_j}$, which depends only on the weights and on $c + c'$.

4.3. The configuration. *Base.* Let $Q = \{3, 6, 10, 11, 16, 18, 20, 21\} \in \mathcal{O}$. There are $1 \cdot 4 \cdot 8 = 32$ sign patterns x on Q with

$$x_3 = +1, \quad x_6 x_{10} x_{18} = -1, \quad x_{11} x_{16} x_{20} x_{21} = +1,$$

and all of them are odd. Delete from B the 31 of them other than $e_3 - e_6 - e_{10} - e_{11} - e_{16} - e_{18} - e_{20} - e_{21}$, which stays. The rest, B' , has $840 + 209 \cdot 128 + 97 = 27689$ points.

Sign group. Let $\Sigma \subseteq C$ be the span of 92361, 20689 and 346115, with supports

$$\{1, 4, 7, 8, 12, 14, 15, 17\}, \quad \{1, 5, 7, 8, 13, 15\}, \quad \{1, 2, 12, 15, 17, 19\}.$$

For $h \in \Sigma$ let S_h change the signs of the coordinates in h . The group $\{S_h\}$ has order 8. The three generators miss Q , so every S_h fixes every vector supported on Q , including the 31 deleted points.

New points. Let $r_1, \dots, r_{384} \in \mathbb{Z}^{21}$ be the vectors in the **representatives** field of **data/D21_30761_certificate.json** in the accompanying repository. They have no zero entries, their entries have absolute values between 149073504 and 249088601, and their squared norms lie between $0.999999998 \cdot 10^{18}$ and $1.000000002 \cdot 10^{18}$. Appendix A gives one of them for each coset. The new points are the 3072 vectors $S_h(r_i)$.

The sign vectors of the new points are exactly the six cosets $c_0 + C$ with

$$c_0 \in \{696, 1993, 2787\} \text{ (odd)}, \quad c_0 \in \{1393, 2139, 3370\} \text{ (even)},$$

with 64 representatives and 512 points on each coset. So the construction drops the odd coset of 3986, which Cohn and Li use, and takes three even cosets instead. On the cosets of 1993, 2139 and 3370 the weights follow the line model, with the lines $\{8, 15, 16, 18, 19\}$, $\{1, 3, 5, 17, 18\}$ and $\{4, 5, 6, 8, 10\}$ respectively: after scaling each representative to $|\lambda|^2 = 18$, every entry is within $1.6 \cdot 10^{-5}$, $1.2 \cdot 10^{-5}$ and $9.4 \cdot 10^{-4}$ of the model. On the other three cosets the weights are irregular, and each representative's entries lie between 0.82 and 1.15 times their root mean square. The weights also vary slightly between the 64 orbits of Σ inside a coset.

We do not have a closed form for the irregular weights. They were found by numerical optimization of the conditions of the previous subsection, with a target cosine slightly below $1/2$, and then rounded to integers at scale about $2 \cdot 10^8$. The proof does not depend on how they were found. Replacing the three line-model cosets by their exact limits would not shorten the proof, because the irregular cosets would still need a computer check.

The total is $27689 + 3072 = 30761$.

4.4. Proof.

Lemma 4.2. *Every S_h , $h \in \Sigma$, maps B' onto itself.*

Proof. Every word of Σ lies in $C \subseteq D$, so it meets every octad of $\mathcal{O}$ in an even number of coordinates. A sign change S_h maps roots to roots. On an octad vector with support o it changes an even number of signs, so the number of minus signs on o stays odd. So $S_h(B) = B$. Since S_h fixes the 31 deleted points, $S_h(B') = B'$. $\square$

Each S_h is orthogonal and $S_a S_b = S_{a+b}$. Hence

$$\langle S_a r_i, S_b r_j \rangle = \langle r_i, S_{a+b} r_j \rangle, \quad \langle S_a r_i, b \rangle = \langle r_i, S_a b \rangle.$$

With Lemma 4.2, every pair involving a new point is determined by one of these finite lists:

- (1) the $\binom{384}{2} \cdot 8 + 384 \cdot 7 = 590976$ products $\langle r_i, S_h r_j \rangle$ with $i \leq j$, excluding $i = j$, $h = 0$;
- (2) the $384 \cdot 27689 = 10632576$ products $\langle r_i, b \rangle$ with $b \in B'$.

Proposition 4.3. *Let $L = 2 \cdot 10^8$. Every product in lists (1) and (2), with x, y the two vectors, satisfies $\langle x, y \rangle \leq (\frac{1}{2} - \frac{1}{L})|x||y|$. In particular every such pair satisfies (1).*

Proof. This is a finite computation in exact integers. For $g = \langle x, y \rangle > 0$ the inequality is equivalent to $(2Lg)^2 \leq (L-2)^2|x|^2|y|^2$. For list (2), Lemma 2.2 gives the maximum of $\langle r_i, b \rangle$ over each full fiber of odd vectors and over the roots in closed form, and the 97 retained points on Q are checked one by one. The program **scripts/verify_d21_orbits.py** carries out exactly this computation with Python integers, in about 2 seconds. A second program, **scripts/check_pairs.py**, written independently, does not use the reduction and tests all 473104180 pairs of the configuration directly, in about one minute. The Lean formalization of Section 5 also checks all pairs directly. $\square$

Base pairs are compatible by Lemma 2.1. So all 30761 points are pairwise compatible, and they span $\mathbb{R}^{21}$ because B' contains all 840 roots. This proves Theorem 1.1 for $n = 21$.

4.5. Remarks. The smallest value of $\frac{1}{4} - \cos^2$ over pairs involving a new point is about $9.34 \cdot 10^{-9}$, attained by two new points whose sign vectors are at Hamming distance 5. The tightest pair of a new point and a base point has $\frac{1}{4} - \cos^2 \approx 1.01 \cdot 10^{-8}$. All 12959199 contacts are between base points.

The deletion is necessary for these representatives. A direct check shows that each of the 31 deleted points has cosine between 0.50276 and 0.50299 with some new point. The octad Q is the sum of the two lines $\{3, 9, 10, 11, 16\}$ and $\{6, 9, 18, 20, 21\}$, and the reduced weights on lines push some new points slightly too close to part of the fiber over Q . The group Σ fixes Q pointwise, so its orbits are free to carry different weights near Q .

With the base held fixed, each new point may be moved independently to any unit vector within chord distance $1/(8 \cdot 10^8)$ of its position, and the result is again a kissing configuration, because each inner product then changes by at most $2/(8 \cdot 10^8) < 1/L$. So with B' fixed these configurations form an open set of dimension $3072 \cdot 20 = 61440$. This freedom does not seem to produce room for a 30762nd point, and we have not tried to prove that it cannot.

5. FORMAL VERIFICATION

We formalized Theorem 1.1 in Lean 4 [4] with Mathlib [9], using Lean 4.33.1 and Mathlib at tag `v4.33.1`. The main theorems `kissing_d19` and `kissing_d21` state, for $(n, N) = (19, 12268)$ and $(21, 30761)$, that there is a finite set S in `EuclideanSpace ℝ (Fin n)` with

$$|S| = N, \quad \|x\| = 1 \quad (x \in S), \quad \langle x, y \rangle \leq \frac{1}{2} \quad (x \neq y \in S).$$

A further lemma restates this in terms of spheres: the unit balls centred at $2x$, $x \in S$, touch the unit ball at the origin, and their centres are at pairwise distance at least 2. We do not define τ_n in Lean. The theorems are the existence statements that the bounds $\tau_n \geq N$ mean.

The development has two layers.

- (1) A plain Lean library, `lean/KissingCheck`, builds each configuration as an array of integer rays from its compact description: Ho's twelve generators and the rules of Section 3, or the Golay polynomial, the deletion rule, the three generators of Σ and the 384 representatives. Each ray is a positive multiple of a point in the text; for instance, the 19-dimensional base rays are $(2b_H, Rb_T)$. One Boolean function tests (1) on every pair, and another tests that every ray has n coordinates and is nonzero.
- (2) A Mathlib library, `lean/KissingD19D21` (the bridge is in `Bridge.lean`), proves, with kernel-checked proofs, that if both tests return true then the normalized rays are N distinct unit vectors with pairwise inner products at most $1/2$. It identifies the Euclidean inner product of two rays with their integer dot product, derives cosine at most $1/2$ from (1), and uses cosine less than 1 to show that the normalized rays are distinct.

The tests are evaluated with `native_decide`. This tactic compiles the Boolean function to native code, runs it, and records the answer through the `Lean.ofReduceBool` mechanism. In this Lean version each use appears in `#print axioms` as its own auxiliary axiom. Each main theorem depends on `propext`, `Classical.choice`, `Quot.sound` and three such axioms, for the size of the array, the nonzero rays and the pairs. Trusting them means trusting that Lean's compiler, its C code generator and its arbitrary-precision integer runtime compute what the definitions say. A kernel evaluation of 473 million pairs is out of reach. The two all-pairs tests take about 6 and 76 seconds of native computation on one core, and the whole project builds in about four minutes on a laptop.

Only the conclusion is formally verified. The theorems depend on the three tests and on the bridge in layer (2), not on the construction code. Lean does not verify that the arrays it builds match the descriptions in Sections 3 and 4, and it does not verify our case analysis. We

checked the match separately. The script `scripts/compare_with_witness.py` reduces each ray built in Lean (written to `lean/out/` by lake exe `dumprays`) and each ray in the files `data/D19_12268_rays.txt` and `data/D21_30761_rays.txt` to its primitive integer vector, and finds the two sets equal, with 12268 and 30761 directions.

The development also contains sanity checks. Kernel-checked `decide` proofs show that the pair test accepts cosine exactly $1/2$ and rejects a pair with $4\langle x, y \rangle^2 - |x|^2|y|^2 = 1$ at size $4 \cdot 10^{18}$. Further checks in `lean/KissingD19D21/Controls.lean`, again by `native_decide`, show that each of the 192 and 31 deleted points conflicts with some added point.

6. CONCLUSION

We have shown that $\tau_{19} \geq 12268$ and $\tau_{21} \geq 30761$, improving the previous lower bounds of 11948 and 29768.

USE OF ARTIFICIAL INTELLIGENCE

GPT-6-Astra Pro and Claude Opus 5.5 were used to find these constructions, write the Lean formalization, and draft this paper.

DATA

The accompanying GitHub repository contains short plain-text descriptions of both constructions (`constructions/`), the coordinates of both configurations and the 384 representatives (`data/`), the verification programs (`scripts/`) and the Lean project (`lean/`).

APPENDIX A. SAMPLE REPRESENTATIVES

Table 5 lists the first representative on each of the six cosets, in the order in which they appear in `data/D21_30761_certificate.json`. Each is written as three rows of seven entries, coordinates 1 to 21.

TABLE 5. One representative r_i per coset of C .

r_1 , coset 696	245706065	220176322	210895444	-204059252	-224413649	-217989311	215994413
	-220176649	217089129	-239789301	240289315	216266534	220176503	204279385
	235748003	200713106	232502335	215134338	225020821	180533251	182339075
r_{65} , coset 1993	-235701710	235701773	235701768	-235701797	235701802	235701767	-235701767
	-149073720	-235701757	-235701769	-235701712	235701769	235701762	235701783
	149073662	149073719	235701771	149073688	149073673	235701774	235701800
r_{129} , coset 2787	-226843642	-181230685	248698137	228776267	191111030	-235154370	-215914780
	-205294560	198569393	-230968394	224770182	-187420697	223040496	210187211
	204943622	248910521	216451072	216471529	222393657	225180933	225249207
r_{193} , coset 1393	-215417013	190119836	232807624	230183890	-199978034	-241264966	-235855976
	206544244	-182467407	215013700	-209982939	184789033	226181509	203403267
	239024609	235071589	224046077	221563853	221563864	229160702	224048159
r_{257} , coset 2139	-149073644	-235701789	149073718	-235701785	-149073714	235701712	-235701783
	235701784	235701792	235701715	235701714	-235701788	235701784	235701787
	235701712	235701788	149073720	149073647	235701785	235701785	235701787
r_{321} , coset 3370	235665358	-235664866	235665258	-149164645	149111025	-149165750	235665338
	149164875	-235665616	149166315	-235719910	-235665327	235719447	235665826
	235719594	235666200	235719560	235718895	235720084	235665599	235665686

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